Hyojun Jeong

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Education

Carnegie Mellon University — Pittsburgh, PA
B.S. in Mechanical Engineering — Aug 2018 - May 2022
M.S. in Mechanical Engineering — Aug 2026- May 2028

Experience

CMU Robomechanics Lab — Pittsburgh, PA
Graduate Research Student — Aug 2026 - May 2028
- Developed pterosaur-inspired adaptive wing mechanisms in SolidWorks, optimizing mechanical design to enable dynamic flight stabilization
- Built high fidelity physics simulations in Isaac Lab, applying reinforcement learning to optimize wing control policies
- Integrated onboard electronics into the glider body within a mass budget, prototyping with 3D printing and machining

General Motors — Warren, MI
Virtual Design Development and Validation Engineer — May 2022 - Aug 2026
- Supported and served as a CAE contact for 2 high-priority programs under accelerated development timelines
- Developed an AI tool with PhysicsAI to predict steering system modal performance, reducing HPC demand and manual effort by 25%; leveraged sensitivity analysis of 42 design variables to enable faster iterations and early-cycle risk detection.
- Led the rapid development of a universal steering wheel armature for cross-platform implementation, using LS-DYNA and Optistruct to analyze structural and durability load cases

CMU Jayan Lab — Pittsburgh, PA
Undergraduate Research Assistant — Jan 2022 - May 2022
- Designed, implemented, and tested fixtures to integrate the spectrophotometer system with a chemical-handling robot using SolidWorks
- Assembled components of a commercial spectrophotometer system to obtain UV-Vis spectra and troubleshooted potential hardware/software related performance issues

Projects

MoonRanger — Jan 2021 - Dec 2021
- Designed a 24lb autonomous lunar rover chassis using carbon fiber to optimize for weight, extreme thermal expansion, and a high stiffness-to-weight ratio
- Validated structural integrity via 20+ stress simulations in ANSYS and Solidworks to ensure survivability during launch vibrations and harsh lunar terrain
- Optimized high-density internal layouts by constructing 3D representations of electronics and sensors within a suitcase-sized footprint

Additional

CAD: SolidWorks, Fusion 360, Siemens NX
CAE: ANSYS, LS-DYNA, Optistruct, ANSA, HyperWorks, nCode
Languages: Python, C++, MATLAB
AI/ML and Robotics: PyTorch, MuJoCo, Isaac Lab